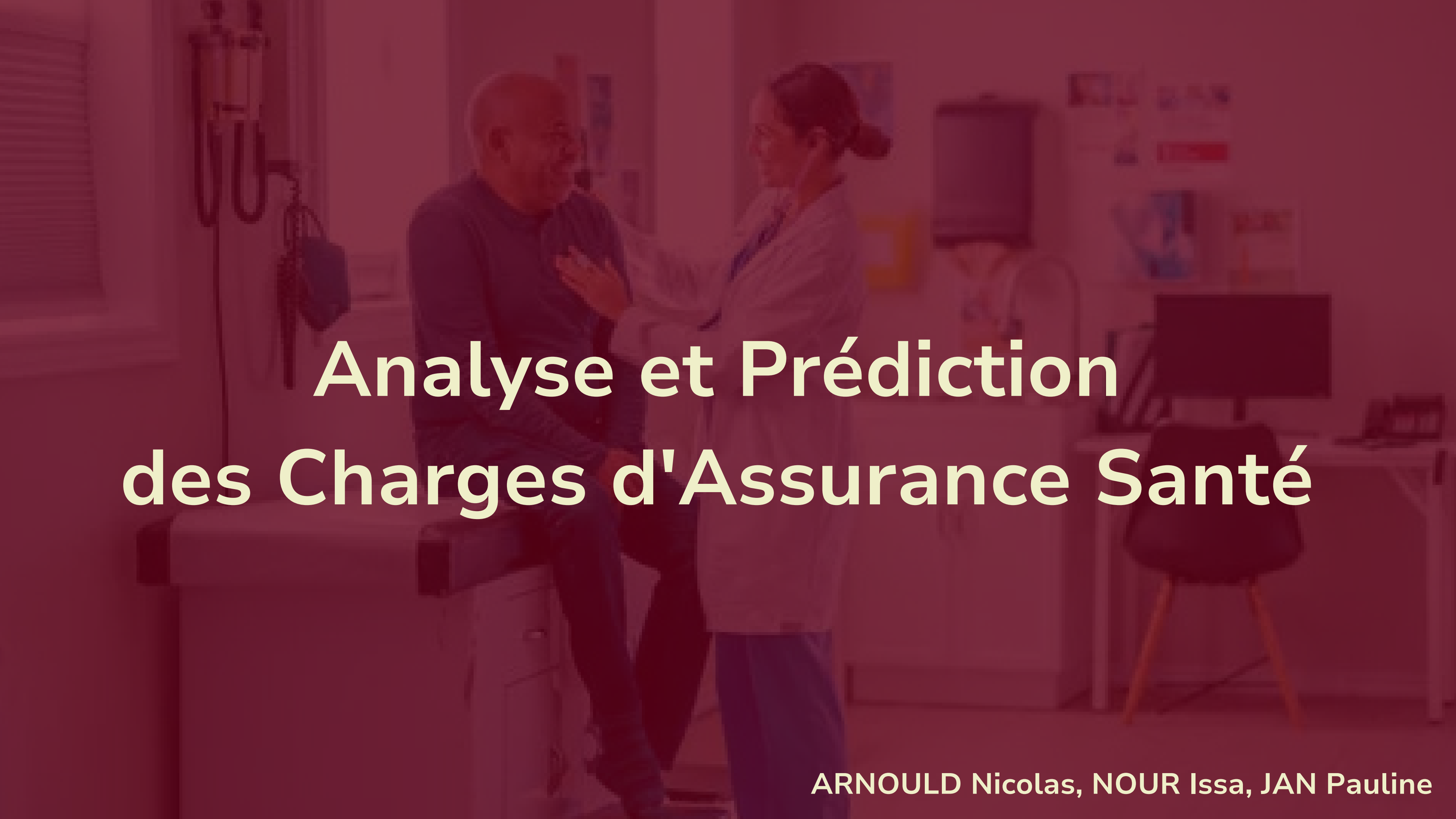


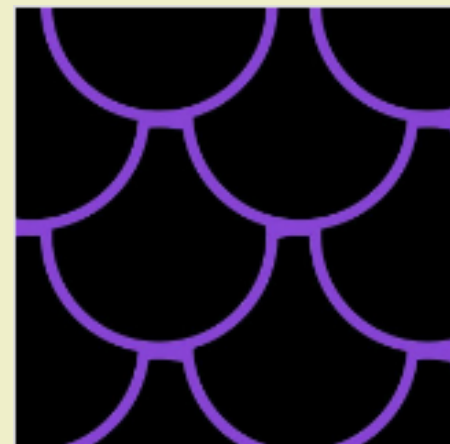
A doctor in a white coat is examining a patient's arm in a clinical setting. The patient is seated, and the doctor is standing, holding the patient's arm. The background shows medical equipment and a desk with a computer monitor.

Analyse et Prédiction des Charges d'Assurance Santé

ARNOULD Nicolas, NOUR Issa, JAN Pauline

Business goal

Comment les caractéristiques individuelles déterminent-elles les coûts de santé ?

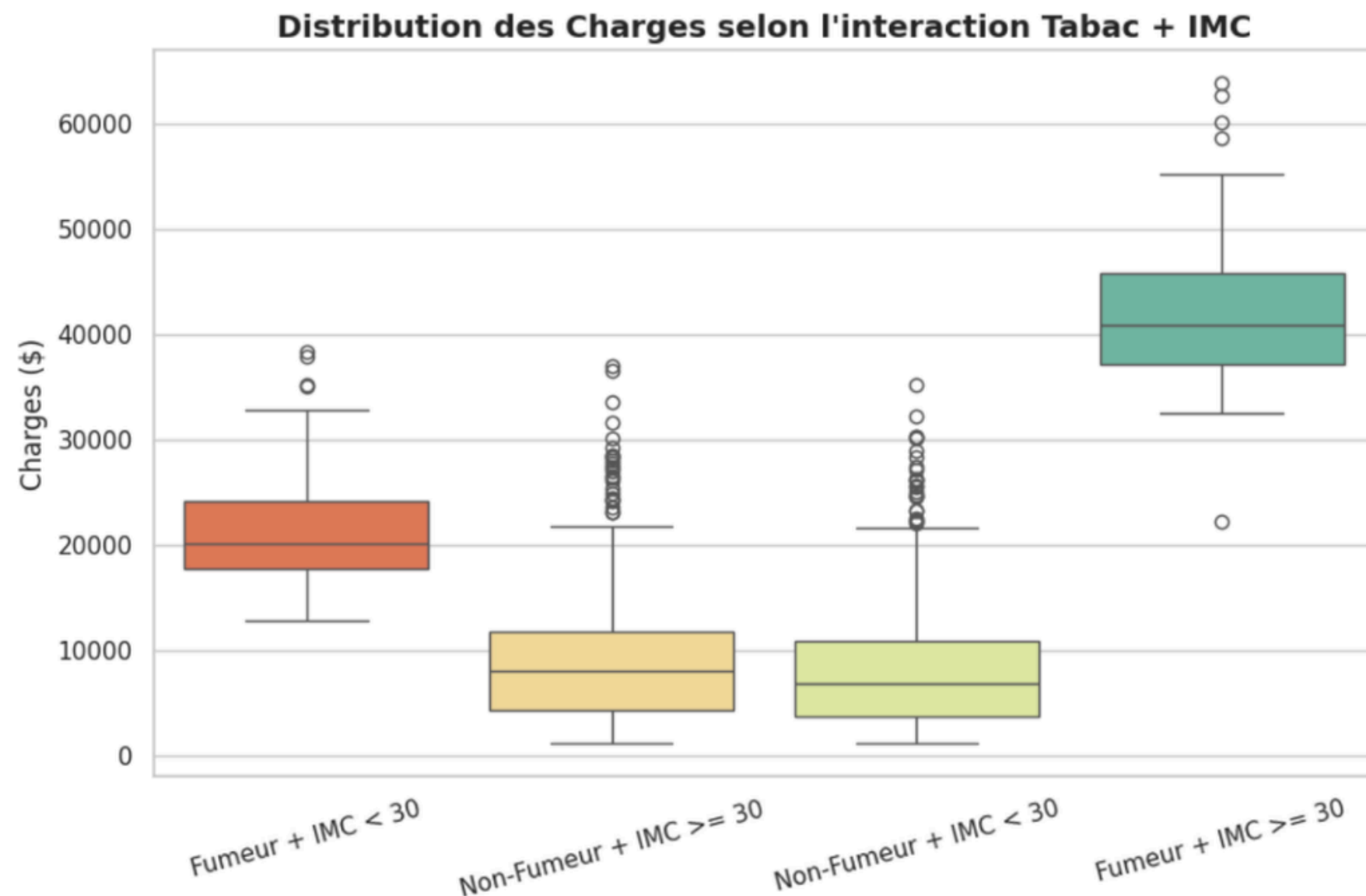


mirichoi0218/insurance

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Data description



Chiffres clés

1337 observations distinctes

7 variables

0 valeur manquante

1 doublon supprimé

Âge/IMC/Enfants

Numérique Explicatif

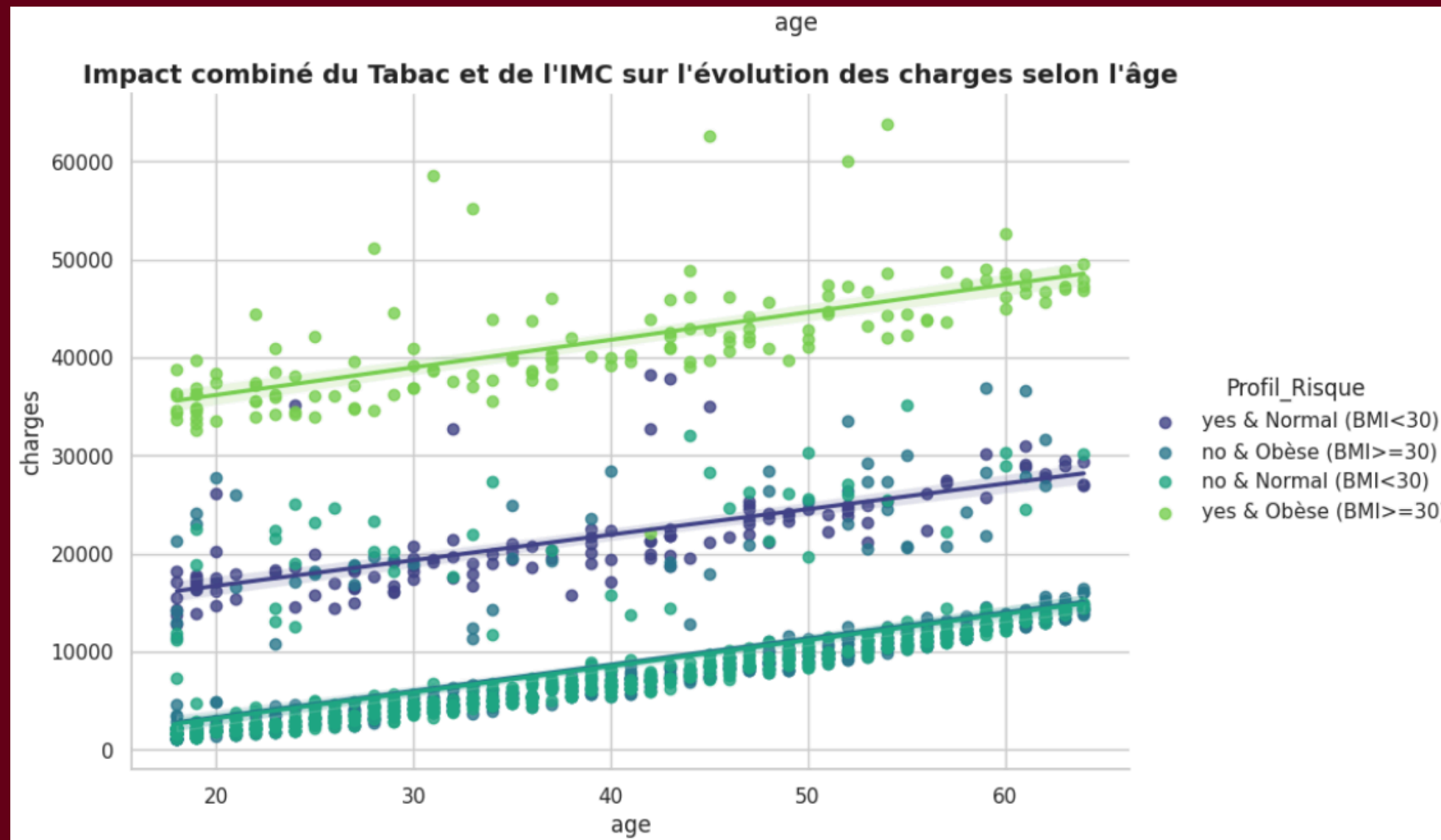
Sexe/Tabac/Région

Catégoriel Explicatif

Charges

Continu Cible

Handcrafted features



Corrélation Tabac seul : 0.7872
Corrélation Profile_Risque_Score : 0.8137

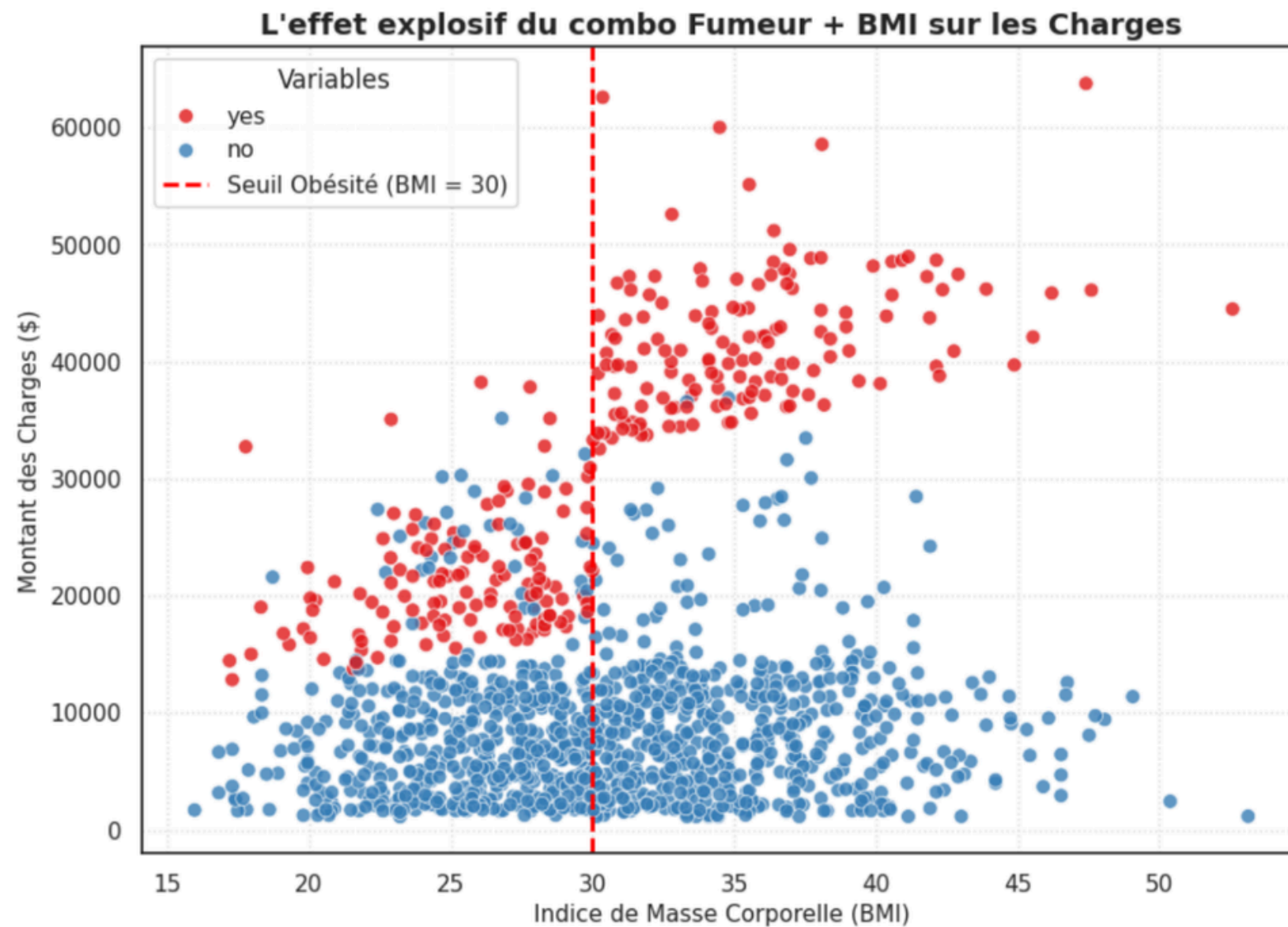
Sexe : Encodé en binaire

Tabac : Encodé en binaire

IMC : Transformé en catégorie binaire (un seuil de 30, définissant l'obésité).

Région : Encodée en 4 catégories distinctes

Regression double



Pour ceux qui fument on ne garde
que l'âge et le bmi

Coefficients:

(Intercept)	X3Smoke	X1Smoke
-22384.8	1440.3	264.9

Pour ceux qui ne fument pas :
le sexe, l'age et le nombre d'enfants.

Coefficients:

(Intercept)	X1	X4	X2
-3348.7	265.3	562.6	465.2

Regression Triple

fumeur + BMI>30

Variable	Coefficient (Poids)
region_northwest	0.016093
bmi	0.013525
age	0.006671
children	0.000904
region_southwest	-0.006814
region_southeast	-0.012990
sex_male	-0.013094

fumeur + BMI<30

Variable	Coefficient (Poids)
bmi	0.023247
children	0.019729
age	0.011642
region_northwest	0.009941
region_southeast	0.007411
region_southwest	0.005096
sex_male	-0.039951

autre

Variable	Coefficient (Poids)
children	0.134965
age	0.045119
bmi	0.000004
region_northwest	-0.047052
sex_male	-0.096517
region_southeast	-0.127507
region_southwest	-0.148254

